

Problem

If $f(\phi) = \cos \phi + j \sin \phi$, show that $f(\phi) = ej\phi$.

Step-by-step solution

Step 1 of 3

Consider the function,

$$f(\phi) = \cos \phi + j \sin \phi \dots\dots (1)$$

Differentiate the equation w.r.t to ϕ .

$$\begin{aligned} \frac{df(\phi)}{d\phi} &= -\sin \phi + j \cos \phi \\ &= j^2 \sin \phi + j \cos \phi \quad (\text{Since } j^2 = -1) \\ &= j(\cos \phi + j \sin \phi) \\ &= jf(\phi) \end{aligned}$$

$$\frac{df(\phi)}{f(\phi)} = jd\phi \dots\dots (2)$$

Step 2 of 3

Integrate equation (2) on both sides.

$$\begin{aligned} \int \frac{df(\phi)}{f(\phi)} &= \int jd\phi \\ \ln[f(\phi)] &= j\phi + \ln(A) \\ \ln[f(\phi)] - \ln(A) &= j\phi \\ \ln\left[\frac{f(\phi)}{A}\right] &= j\phi \\ \frac{f(\phi)}{A} &= e^{j\phi} \\ f(\phi) &= Ae^{j\phi} \dots\dots (3) \end{aligned}$$

Step 3 of 3

Equate equation (1) and equation (3).

$$Ae^{j\phi} = \cos \phi + j \sin \phi$$

Put 0 for ϕ in the equation.

$$Ae^{j(0)} = \cos(0) + j \sin(0)$$

$$A = 1$$

Substitute 0 for A in equation (3).

$$\begin{aligned} f(\phi) &= (1)e^{j\phi} \\ &= e^{j\phi} \end{aligned}$$

Therefore, $f(\phi) = e^{j\phi}$.

Hence, the equation is proved.